

Michail Mitsakis

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Education

- MSc** [Technical University of Denmark (DTU)], Materials Science [2021 – 2023]
- Thesis: optimisation of electrodeposition parameters for hydrogen evolution reaction (HER) catalysts.
 - Specialisation: electrochemistry and green hydrogen.
- BSc** [National and Kapodistrian University of Athens (NKUA)], Physics [2015 – 2020]

Experience

- [WMG Sustainable Innovation]**, [Energy Consultant] Greece [2024] – present
- Client-facing delivery on deep-tech and EU-funded programmes in hydrogen technologies, manufacturing data spaces, and shipyard digitalisation.
 - Proposal development, technical scoping, and meeting preparation with industrial and research partners.

Projects

- catalyst-kg-agent** Jan 2026
- Knowledge-graph-grounded, cost-aware multi-agent system for materials-discovery campaigns.
- Decides between a knowledge-graph lookup, an MLIP surrogate query (MACE), and an expensive simulated experiment under an explicit budget, as in a self-driving-lab workflow.
 - Materials Project knowledge graph of 189 HER/OER-relevant materials; zero-shot MACE reaches 0.114 eV/atom MAE on non-oxides.
 - CGCNN baseline evaluated with composition-disjoint cross-validation to control polymorph leakage.
 - Stack: PyTorch Geometric, MACE, BoTorch/Ax, pydantic-ai, Ollama, NetworkX, MLflow.
- notion-second-brain** Jan 2026
- Fully local retrieval-augmented agent over a personal Notion workspace.
- Hybrid dense + BM25 retrieval with reciprocal-rank fusion and cross-encoder reranking.
 - Persistent memory, OCR ingestion of PDFs and images, and an anchored-rubric evaluation harness.
 - Stack: pydantic-ai, Ollama, Qdrant, Streamlit, Docker.
- CaMEL-RAG — LLM Hackathon for Materials Science & Chemistry** Jan 2025
- Retrieval-augmented LLM for natural-language queries over Open Catalyst adsorption-energy data (team Code4Catalysis-KFUPM).
- FAISS vector store over sentence-transformer embeddings; answers cite the retrieved records they are grounded in.
 - Returned stored DFT adsorption energies exactly on 500 in-index test queries ($R^2 = 1.00$), validating retrieval and grounding.
 - Stack: Python, FAISS, sentence-transformers, OpenAI API.

BayBE one more time — Bayesian Optimization Hackathon for Chemistry and Materials

Jan 2024

Bayesian optimisation with BayBE to screen small-molecule corrosion inhibitors for aluminium alloys (team Surface Science Syndicate).

- Compared one-hot, Mordred, RDKit and Morgan-fingerprint encodings against random search in simulated 50-experiment campaigns (10 Monte Carlo repeats).
- Transfer learning from AA1000 data reached ~96% inhibition efficiency on AA2024 by the 12th experiment, versus ~89% after 25 without it.
- Stack: BayBE, RDKit, Mordred, pandas.

Publications

From Knowledge to Action: Outcomes of the 2025 Large Language Model (LLM) Hackathon for Applications in Materials Science and Chemistry

May 2026

Community paper; contributed CaMEL-RAG (team Code4Catalysis-KFUPM).

A. Roy, K. Shen, A. MacBride, ..., **M. Mitsakis**, ... (300+ authors)

arxiv.org/abs/2605.03205 (arXiv preprint arXiv:2605.03205)

Bayesian Optimization Hackathon for Chemistry and Materials

June 2025

Community paper; contributed BayBE corrosion-inhibitor study (team Surface Science Syndicate).

S. Baird, M. Ansari, Z. Afzal, ..., **M. Mitsakis**, ... (125 authors)

doi.org/10.26434/chemrxiv-2025-dzh5z (ChemRxiv)

Skills

Electrochemistry

Electrocatalysis, hydrogen evolution reaction, electrodeposition, cyclic voltammetry, impedance spectroscopy, ECSA analysis

Machine learning for materials

Bayesian and multi-objective optimisation (BayBE, BoTorch/Ax, Dragonfly), graph neural networks, machine-learned interatomic potentials (MACE), molecular featurisation (RDKit, Mordred), Materials Project, Open Catalyst

LLMs and agents

RAG (FAISS, Qdrant, hybrid retrieval, reranking), multi-agent systems (pydantic-ai), local inference (Ollama), evaluation harnesses

Software

Python, PyTorch Geometric, pandas, MLflow, Docker, GitHub Actions, Streamlit

Project delivery

EU-funded project proposals, technical scoping, consortium work